Short-Term Market Changes and Market Making with Inventory*

Seyoung Park†, Bong-Gyu Jang**,§, Jin Gi Kim**, Ho-Seok Lee‡, and Sam Beatson†

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§ Corresponding Author

† Nottingham University Business School, University of Nottingham, United Kingdom,

E-mail: seyoung.park@nottingham.ac.uk (Park) and liasb@nottingham.ac.uk (Beatson)

** Department of Industrial and Management Engineering, POSTECH, Republic of Korea, E-mail: bonggyujang@postech.ac.kr (Jang) and dgkimjg0203@postech.ac.kr (Kim)

‡ Department of Mathematics, Kwangwoon University, Republic of Korea, Email: hoseoklee@kw.ac.kr

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ABSTRACT

We examine the optimal trading strategy of a market maker managing stock inventory under short-term market fluctuations, particularly shifts in trading intensity and volatility. To capture these dynamics, we model market condition changes using Poisson jump processes. Within this framework, we derive closed-form solutions for the market maker's optimal bid and ask quotes and analyze the resulting inventory dynamics. Our findings highlight how transient changes in market activity shape optimal quoting behavior and inventory management.

Keywords: market microstructure, regime switching, market making, inventory management, bid-ask spread

JEL classification: C61, D40, G11

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1 Introduction

Dealers in securities markets play a fundamental role in providing liquidity and enabling price discovery. By absorbing customer demand and supply, they allow buyers and sellers to transact efficiently. Traditionally, this role was fulfilled by market-makers or specialists who stood ready to buy or sell from their own accounts. In modern markets, however, electronic communications networks and algorithmic trading systems have transformed liquidity provision. Limit order traders—who post bids and asks at prices relative to the prevailing mid-quote—can now compete directly with dealers. Such traders may act passively as liquidity suppliers or actively as aggressors through market orders. The transparency of the limit order book, observable through depth-of-market (DOM) feeds, incentivizes algorithmic strategies that continuously reassess whether to submit limit or market orders based on perceived price efficiency and spread conditions.

The economics of the bid–ask spread remain central to understanding these behaviors. A wide spread increases the cost of aggressive orders and encourages traders to supply liquidity with limit orders, while a narrow spread reduces trading costs and makes market orders more attractive. In either case, liquidity providers earn profits from spread capture but face risks from uncertain execution and adverse selection.

The academic study of optimal dealer behavior in such settings has a long tradition. Ho and Stoll (1981) modeled a single dealer's optimal bid and ask quotes using dynamic programming to maximize terminal wealth. Avellaneda and Stoikov (2008) extended this framework by deriving closed-form solutions for optimal spreads under constant absolute risk aversion (CARA) utility. While these models provide foundational insights, they assume constant investment opportunities and abstract away from stochastic variation in market conditions such as time-varying trading intensity and volatility.

Empirical research shows, however, that market activity is far from constant. Easley et al. (2008) demonstrate that arrival rates of informed and uninformed trades vary over time, while Chordia et al. (2001) highlight the volatility of daily trading activity. Intraday studies also reveal systematic patterns in trading volume, with peaks at the open and close of trading sessions (Abhyankar et al. 1997; Hamao and Hasbrouck 1995). Moreover, a robust body of evidence links trading volume and price volatility (Bessembinder 1994; Darrat et al. 2003, 2007). Together, these findings suggest that realistic models of dealer behavior must account for stochastic and state-dependent market conditions.

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More specifically, trading intensity encompasses both the frequency of transactions—how often trades are executed or filled—and the trading volume, defined as the cumulative size of positions or contracts exchanged within a given time window. In high-frequency trading environments, observable through the limit order book near the equilibrium price, trading intensity can thus be inferred from transaction volumes over short intervals (Gopikrishnan et al. 2000; Maslow and Mills 2001; Gabaix et al. 2006; Weber and Rosenow 2005; Potters and Bouchaud 2003). On the other hand, a large body of research in market microstructure has established a strong contemporaneous link between trading volume and price volatility. Early studies, including Bessembinder (1994), Galati (2000), and Hartmann (1999), document a positive relationship between unexpected trading volume and volatility. More recent work has reinforced these findings: Darrat et al. (2003) show that trading volume and return volatility are positively correlated, while Darrat et al. (2007) and Floros and Vougas (2007) examine this relationship at the daily frequency. Building on this literature, our paper models the positive association as simultaneous shifts in both trading volume and volatility, captured through the Poisson jump process introduced earlier.

In this paper, we extend the utility-maximizing approach of Ho and Stoll (1981) and Avellaneda and Stoikov (2008) by incorporating short-term market changes into the dealer's problem. Specifically, we model trading intensity and volatility as switching between high and low states via a Poisson jump process. This framework captures the empirical simultaneity of shifts in order flow and volatility documented in the microstructure literature. Our analysis yields two main results. First, market makers alter their inventory management behavior over the trading day: early in the session they prioritize limiting risk exposure, whereas near the close they shift focus toward maximizing terminal wealth. Second, we find that dealers who account for short-term fluctuations in market conditions optimally accept greater inventory risk.

Our contribution lies in bridging the gap between classical models of dealer optimization and the empirical reality of fluctuating trading intensity and volatility. By explicitly modeling short-term market changes, we provide a framework that better aligns with the dynamics observed in modern limit order markets and high-frequency environments.

The paper proceeds as follows. Section 2 specifies our model set-up, and Section 3 derives optimal strategies of a market making agent dealing in conditions of short-term market changes where both intensity and volatility may alter together. In Section 4, we

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conduct an extensive numerical analysis. Finally, Section 5 concludes.

2 The Basic Model

We consider a market makers of a stock, who is a monopolistic agent, and she is willing to trade shares of the stock through limit orders to make profit and can trade two financial assets: a bond (or a money market instrument) and the stock. We assume that she plans and develops her bidding and asking strategies on a short-term (or daily) basis, thus, zero return of the money market instrument is considered.

Following the observations of Abhyankar *et al.* (1997), Hamao and Hasbrouck (1995) and others we take the stock market to be a stochastically changing situation. More specifically, its trading intensity (or equivalently, trading frequency and volume) and volatility changes drastically even for a short time period. This could be modeled by exploiting a Markov regime-switching process; we employ two regimes, a high-volatility regime (or Regime H) and a low-volatility regime (or Regime L), and Regime $i \in \{H, L\}$ shifts into Regime j ($j \neq i,\ j \in \{H, L\}$) at the first jump time of a non-homogeneous Poisson process, call $N_{i,t}$, with intensity $\lambda_{i,t}$.

The mid-market stock price y_t in Regime i evolves according to

$$dy_t = \sigma_i dW_t,$$

where W_t is a standard Brownian motion, and σ_i, $i \in \{H, L\}$, are positive constants such that $\sigma_H > \sigma_L$. Denote by L_t (U_t) the cumulative shares purchased (sold, respectively) up to the time t. Then the dollar amount x_t of the market makers' cash is given by

$$dx_t = -(y_t - B_t^i)dL_t + (y_t + A_t^i)dU_t,$$

where B_t^i (A_t^i) represents the distance between the mid-market stock price y_t and the market makers' bid price (ask price, respectively) in Regime i. They satisfy

$$\begin{cases} B_t^i = y_t - bid_t^i, \\ A_t^i = ask_t^i - y_t, \end{cases}$$

where bid_t^i and ask_t^i denote the bid and ask prices of the market makers in Regime i, respectively. We model L_t and U_t as non-homogeneous Poisson processes with intensities $I^i(B_t^i)$, $I^i(A_t^i)$, respectively. Therefore, we have

$$dL_t = \begin{cases} 1 & \text{with probability} \quad I^i(B_t^i)dt \\ 0 & \text{with probability} \quad 1 - I^i(B_t^i)dt \end{cases}$$

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and

$$dU_t = \begin{cases} 1 & \text{with probability} \quad I^i(A_t^i)dt \\ 0 & \text{with probability} \quad 1 - I^i(A_t^i)dt \end{cases}$$

$I^i(B_t^i)$ and $I^i(A_t^i)$ represent trading intensities of market sell and buy orders in Regime i, and they changes according to bid-ask spreads. We assume that there is a market risk source W_t and non-homogenous Poisson processes $N_{i,t}$, L_t, U_t are all mutually independent. Following Avellaneda and Stoikov (2008), we take the following trading (or order arrival) intensities across the two regimes:

$$I^i(A_t^i) = C_i \exp(-k_i A_t^i),$$
$$I^i(B_t^i) = C_i \exp(-k_i B_t^i),$$

where C_i's and k_i's are positive constants. [1] Notice that in our model we are assuming both the trading intensity $I^i(\cdot)$ with C_i and k_i and volatility σ_i shifts into the trading intensity $I^j(\cdot)$ with C_j and k_j and volatility σ_j at the first jump time of a Poisson process with intensity $\lambda_{i,t}$, for $i, j \in \{H, L\}$. Under this setup, the change in the number of the holding stocks (or inventory) q_t of the market makers satisfy.

$$dq_t = dL_t - dU_t.$$

An market makers' aim is to maximize the expected constant absolute risk aversion (CARA) utility of terminal wealth. As Guéant et al. (2013), the maximum and minimum inventories of the market maker are assumed to be $Q > 0$, $-Q$, respectively. In intensity and volatility i, the market makers' objective is given by the following value function:

$$V^i(x, y, q, t) = \max_{A_t^i, B_t^i} E\Big[-\exp\big(-\gamma(x_T + q_T y_T)\big)\Big], \quad \text{such that} \quad -Q \le q_t \le Q.$$

[1] Avellaneda and Stoikov (2008) let $C_i = \Lambda_i/\alpha_i$ and $k_i = \alpha_i K_i$. Here, Λ_i denotes the frequency of market buy (or equivalently, sell) orders in Regime i which is

$$\frac{\text{total trading volume}}{\text{average size of market orders}}$$

over some time period in Regime i, and K_i represents the proportional relationship between the stock price change Δy and a logarithm of the market order size Q (Potters and Bouchaud, 2003) in Regime i, i.e.,

$$K_i = \frac{\Delta y}{\ln Q}.$$

α_i is the constant concerning the power law of the distribution of the size of market orders in Gopikrishnan *et al. (2000)*:

$$\text{the density of market order size in Regime } i \quad \propto \quad (\text{market order size})^{-1-\alpha_i}.$$

Notice that a higher α_i yields a higher k_i and a lower C_i, thus subsequently, a lower intensity $I^i(\cdot)$.

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where γ is the absolute risk aversion coefficient, x is the initial dollar amount in the bond, y is the initial stock price, and q is the initial number of the holding stocks (or stock inventory) of the market makers. The maximum and minimum inventories of market makers are assumed to be Q, $-Q$.

3 Market Making with Short Term Market Changes

First, we separate the value function V_i $(i = H, L)$ into V_H for the case in which the regime is a high-volatility regime and V_L for the case in which the regime is a low-volatility regime. We name bid and ask distance for regime i $(i = H, L)$ as B_i and A_i, respectively. Using dynamic programming principles again, we derive two Hamilton-Jacobi-Bellman equations systems for this problem:

For $q = Q$:

$$V_t^i + \frac{1}{2}\sigma_i^2 V_{yy}^i + \max_{A_t^i} C_i \exp(-k_i A_t^i)\left[V^i(x + y + A_t^i, y, q-1, t) - V^i(x, y, q, t)\right] + \lambda_{i,t}(V^j - V^i) = 0, \tag{1}$$

For $|q| < Q$:

$$\begin{aligned} &V_t^i + \frac{1}{2}\sigma_i^2 V_{yy}^i + \max_{B_t^i} C_i \exp(-k_i B_t^i)\left[V^i(x - y + B_t^i, y, q+1, t) - V^i(x, y, q, t)\right] \\ &+ \max_{A_t^i} C_i \exp(-k_i A_t^i)\left[V^i(x + y + A_t^i, y, q-1, t) - V^i(x, y, q, t)\right] + \lambda_{i,t}(V^j - V^i) = 0, \end{aligned} \tag{2}$$

For $q = -Q$:

$$V_t^i + \frac{1}{2}\sigma_i^2 V_{yy}^i + \max_{B_t^i} C_i \exp(-k_i B_t^i)\left[V^i(x - y + B_t^i, y, q+1, t) - V^i(x, y, q, t)\right] + \lambda_{i,t}(V^j - V^i) = 0, \tag{3}$$

subject to the terminal condition

$$V^i(x, y, q, T) = -\exp\big(-\gamma(x + qy)\big), \quad \forall q \in \{-Q, \cdots, Q\}. \tag{4}$$

Here, the subscripts stand for derivatives, i.e., $V_t^i = \partial V^i/\partial t$, $V_{yy}^i = \partial^2 V^i/\partial y^2$. Not surprisingly, the last term of the left hand side of equations (1), (2), (3) involves the possibility that the market conditions in Regime j might affect the optimal behaviors of the market makers in Regime i.

We conjecture that the value function has the following form:

$$V^i(x, y, q, t) = -\exp\big(-\gamma(x + qy)\big) \cdot \big\{u_q^i(t)\big\}^{-\gamma/k_i}.$$

The optimal bid distance, the optimal ask distances, and the optimal bid-ask spread are obtained by solving the non-linear PDE (1), (2), (3) and substituting the solution

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into the explicit equations (5), (6), (7). From the first-order optimality condition with respect to B_t^i, we can obtain the explicit equation of the optimal bid distance B_t^i by

$$B_t^i = \frac{1}{\gamma}\log\left(1+\frac{\gamma}{k_i}\right) + \frac{1}{k_i}\log\left(\frac{u_q^i(t)}{u_{q+1}^i(t)}\right) \tag{5}$$

Similarly, the optimal ask distance A_t^i is explicitly given by

$$A_t^i = \frac{1}{\gamma}\log\left(1+\frac{\gamma}{k_i}\right) + \frac{1}{k_i}\log\left(\frac{u_q^i(t)}{u_{q-1}^i(t)}\right) \tag{6}$$

and the optimal bid-ask spread $\delta_t^i = B_t^i + A_t^i$ is given by

$$\delta_t^i = \frac{2}{\gamma}\log\left(1+\frac{\gamma}{k_i}\right) + \frac{1}{k_i}\log\left(\frac{\left(u_q^i(t)\right)^2}{u_{q+1}^i(t)u_{q-1}^i(t)}\right) \tag{7}$$

Putting (5), (6) into the non-linear PDE (1), (2), (3) gives

For $q = Q$:

$$\dot{u}_q^i(t) - \left(\lambda_i - \frac{\gamma^2\sigma^2Q^2}{2}\right)u_q^i(t) - C_i\frac{\gamma}{k_i}\left(1+\frac{\gamma}{k_i}\right)^{-1-k_i/\gamma}u_{q-1}^i(t) + \lambda_{i,t}u_q^j(t) = 0 \tag{8}$$

For $|q| < Q$:

$$\dot{u}_q^i(t) - \left(\lambda_i - \frac{\gamma^2\sigma^2Q^2}{2}\right)u_q^i(t) + C_i\frac{\gamma}{k_i}\left(1+\frac{\gamma}{k_i}\right)^{-1-k_i/\gamma}\left(u_{q+1}^i + u_{q-1}^i(t)\right) + \lambda_{i,t}u_q^j(t) = 0 \tag{9}$$

For $q = -Q$:

$$\dot{u}_q^i(t) - \left(\lambda_i - \frac{\gamma^2\sigma^2Q^2}{2}\right)u_q^i(t) + C_i\frac{\gamma}{k_i}\left(1+\frac{\gamma}{k_i}\right)^{-1-k_i/\gamma}u_{q+1}^i(t) + \lambda_{i,t}u_q^j(t) = 0 \tag{10}$$

subject to the terminal condition

$$u_q(T) = 1, \quad \forall q \in \{-Q, \cdots, Q\}$$

We set new variable $X_i(q)$, Y_i, Z_i for simplicity:

$$X_i(q) = -\frac{\gamma k_i\sigma_i^2}{2}q^2, \qquad Y_i = C_i\left(1+\frac{\gamma}{k_i}\right)^{-1-k_i/\gamma}, \qquad Z_{i,t} = \frac{k_i\lambda_{i,t}}{\gamma}$$

We develop a numerical method to solve our problem. We approximate the equations (8), (9), (10) by utilizing the forward differences for time variable t.

Then, for $q = Q$:

$$u_q^i(t-\Delta t) = \left[1 + \Delta t X_i(q) + \Delta t Z_{i,t} - \Delta t Z_{i,t}\left(\frac{\left\{u_q^j(t)\right\}^{-\gamma/k_j}}{\left\{u_q^i(t)\right\}^{-\gamma/k_i}}\right)\right]u_q^i(t) + \Delta t Y_i u_{q-1}^i(t) \tag{11}$$

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For $|q| < Q$:

$$u_q^i(t-\Delta t) = \Delta t Y_i u_{q+1}^i(t) + \left[1 + \Delta t X_i(q) + \Delta t Z_{i,t} - \Delta t Z_{i,t}\left(\frac{\left\{u_q^j(t)\right\}^{-\gamma/k_j}}{\left\{u_q^i(t)\right\}^{-\gamma/k_i}}\right)\right] u_q^i(t) + \Delta t Y_i u_{q-1}^i(t) \tag{12}$$

For $q = -Q$:

$$u_q^i(t-\Delta t) = \Delta t Y_i u_{q+1}^i(t) + \left[1 + \Delta t X_i(q) + \Delta t Z_{i,t} - \Delta t Z_{i,t}\left(\frac{\left\{u_q^j(t)\right\}^{-\gamma/k_j}}{\left\{u_q^i(t)\right\}^{-\gamma/k_i}}\right)\right] u_q^i(t) \tag{13}$$

subject to the terminal condition

$$u_q(T) = 1, \quad \forall q \in \{-Q, \cdots, Q\}$$

Then, the equations (11), (12), (13) can be solved from backward.

4 Numerical Analysis

This section deals with an extensive numerical analysis. First, we analyze the results of the optimal bid and ask distance. Second, we compare the behavior of the market makers in the single- and two-regime cases. The single-regime denotes the state without short-term market changes. The result of single-regime is the expansion from Avellaneda and Stoikov (2008) to limited inventory.

For single- and two-regime cases, we set the common baseline parameters values are as follows : $Q = 25$, $T = 1$, $\gamma = 0.1$, $\Delta t = 1/1000$. We set the regime switching intensities λ_H, λ_L are as follow:

$$\lambda_{H,t} = \begin{cases} 30, & \text{if} \quad 0 \le t \le 0.9 \\ 1, & \text{if} \quad 0.9 < t \le 1 \end{cases} \qquad \lambda_{L,t} = \begin{cases} 1, & \text{if} \quad 0 \le t \le 0.9 \\ 30, & \text{if} \quad 0.9 < t \le 1 \end{cases}$$

Table 1 displays the baseline parameters under the single- and two-regime cases. For the purpose of comparison, we take average parameters of two-regime model as the parameters for the single-regime model. As a result of running 100,000 simulations with a given regime switching intensities, the average time that the market is in high intensity and volatility is 0.13. Therefore, we assume that parameters in the single-retime model are as follow:

$$\text{(parameter in the single-regime model)} =$$

$$0.13 \times \text{(parameter in Regime H)} + 0.87 \times \text{(parameter in Regime L)}$$

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Table 1: Baseline Parameters

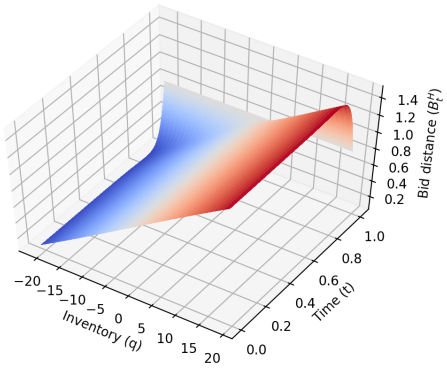
Panel A: Single-regime		Panel B: Two-regime	
c	395.5	c_H	700
		c_L	350
k	1.87	k_H	1.0
		k_L	2.0
σ	2.26	σ_H	4.0
		σ_L	2.0

Following Hamao and Hasbrouck (1995), we assume the market starts in a high intensity and volatile state. Figure 1 shows the behavior with time and inventory of the optimal bid and ask distance. The optimal bid and ask distance heavily depend on the inventory q at the beginning of the trading session since market makers are reluctant to hold a lot of inventory. In our problem, market makers aim to maximize the sum of total assets, that is, dollar amounts and stock values. The more stocks held at the beginning of the trading session, the more market makers are exposed to market risk. As a result, as a market makers' inventory q increases, the optimal bid(ask) distance increases(decreases). It implies that the market makers decide the optimal bid and ask distance in the direction of reducing inventory. However, the optimal bid and ask distance are converged to a certain level without being significantly affected by the inventory q. This means that as the market approaches the end of the trading session, the market maker's risk exposure is reduced, so market makers do not need to reduce their inventory while taking losses.

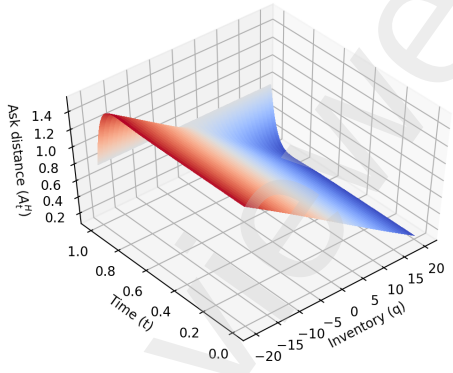
We then run 1,000,000 simulations to compare the inventory of market maker in a two-regime scenario to the inventory of market maker in single-regime scenario. Figure 2 displays standard deviation of inventory for the single- and two-regime case. Figure 2 Panel A shows volatility of inventory over time. The trading session starts with high intensity and volatility, and the probability of closing a transaction is high. Thus, the standard deviation of the market maker's inventory in the two-regime scenario rises faster than in the single-regime scenario. The standard deviation of inventory, which was steadily high during the trading session, is resolved due to high intensity and volatility at the end of the trading session. Because of this, the standard deviation of daily inventory is generally higher in the two-regime compared to the single-regime situation in Figure

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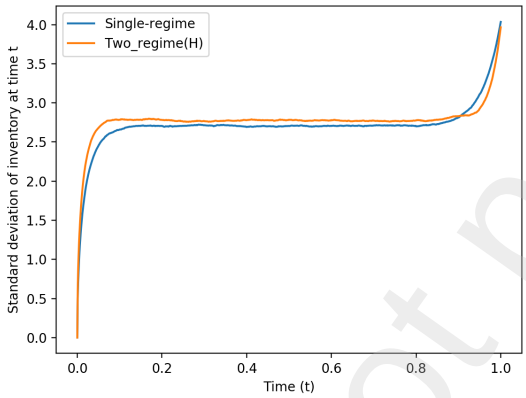


Panel A: The optimal bid distance

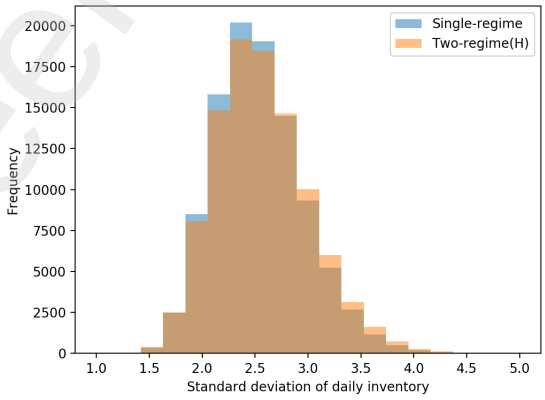


Panel B: The optimal ask distance

Figure 1: The optimal bid and ask distance



Panel A: Standard deviation of inventory at time t



Panel B: Histogram of standard deviation of daily inventory

Figure 2: Standard deviation of inventory

2 panel B. Higher volumes in a particular direction from volatile inventory (e.g. more aggressive buyers than sellers) can cause imbalance of the demand-supply relationship in securities and buying pressure. In the high demand imbalance situation, buyers are willing to buy a security (aggressively) at higher prices (willing to absorb the risk of the spread), while sellers are thereby incentivised to back off (i.e. move their limit orders higher) in order to be able to sell at a higher price if the buying bulls seem to be in charge. If there are insufficient sellers to meet buyers demand at a given price, the effect is that the incremental price of the security increases. Therefore, market makers considering short-term market change tends to take more inventory risk.

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5 Conclusion

This paper investigates the optimal behavior of a market maker in an environment where trading intensities and volatilities are driven by Poisson jump processes. We derive closed-form solutions for the market maker's optimal bid and ask strategies under these conditions. Our analysis shows that short-term market changes have a significant impact on optimal behavior. In particular, at the beginning of the trading session, the market maker prioritizes maximizing total asset value rather than managing inventory. Furthermore, compared with the benchmark case that ignores short-term fluctuations, our model highlights a distinct feature: market makers optimally accept greater inventory risk when accounting for such changes.

By explicitly incorporating stochastic shifts in trading intensity and volatility, this paper extends the classical dealer models and provides new insights into liquidity provision under dynamic market conditions.

References

Abhyankar, A., Ghosh, D., Levin, E., Limmack, R. J., 1997. Bid-ask spreads, trading volume and volatility: Intra-day evidence from the London Stock Exchange. J. Bus. Finance Account. 24(3), 343-362.

Avellaneda, M., Stoikov, S., 2008. High-frequency trading in a limit order book. Quant. Finance 8(3), 217-224.

Bessembinder, H., 1994. Bid-ask spreads in the interbank foreign exchange markets. J. Financ. Econ. 35(3), 317-348.

Chordia, T., Roll, R., Subrahmanyam, A., 2001. Market liquidity and trading activity. J. Finance 56(2), 501-530.

Darrat, A. F., Rahman, S., Zhong, M., 2003. Intraday trading volume and return volatility of the DJIA stocks: A note. J. Bank. Finance 27(10), 2035-2043.

Darrat, A. F., Zhong, M., Cheng, L. T., 2007. Intraday volume and volatility relations with and without public news. J. Bank. Finance 31(9), 2711-2729.

Easley, D., Engle, R. F., O'Hara, M., Wu, L., 2008. Time-varying arrival rates of informed and uninformed trades. J. Financ. Econom. 6(2), 171-207.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5517658

Floros, C., Vougas, D., 2007. Trading volume and returns relationship in Greek stock index futures market: GARCH vs. GMM. Int. Res. J. Finance Econom. 12, 98-115.

Gabaix, X., Gopikrishnan, P., Plerou, V., Stanley, H. E., 2006. Institutional investors and stock market volatility. Q. J. Econ. 121(2), 461-504.

Galati, G., 2000. Trading volumes, volatility and spreads in foreign exchange markets: evidence from emerging market countries. Bank Int. Settl. www.bid.org.

Gopikrishnan, P., Plerou, V., Gabaix, X., Stanley, H. E., 2000. Statistical properties of share volume traded in financial markets. Phys. Rev. E 62(4), R4493-R4496.

Guéant, O., Lahale, C. A., Fernandex-Tapia, J., 2013. Dealing with the inventory risk: a solution to the market making problem. Math. Financ. Econ. 7, 477-507.

Hamao, Y., Hasbrouck, J., 1995. Securities trading in the absence of dealers: Trades and quotes on the Tokyo Stock Exchange. Rev. Financ. Stud. 8(3), 849-878.

Hartmann, P., 1999. Trading volumes and transaction costs in the foreign exchange market: evidence from daily dollar–yen spot data. J. Bank. Finance 23(5), 801-824.

Ho, T., Stoll, H. R., 1981. Optimal dealer pricing under transactions and return uncertainty. J. Financ. Econ. 9(1), 47-73.

Maslov, S., Mills, M., 2001. Price fluctuations from the order book perspective—empirical facts and a simple model. Physica A: Statistical Mechanics and its Applications 299(1-2), 234-246.

Potters, M., Bouchaud, J. P., 2003. More statistical properties of order books and price impact. Physica A: Statistical Mechanics and its Applications 324(1-2), 133-140.

Weber, P., Rosenow, B., 2005. Order book approach to price impact. Quant. Finance 5(4), 357-364.

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